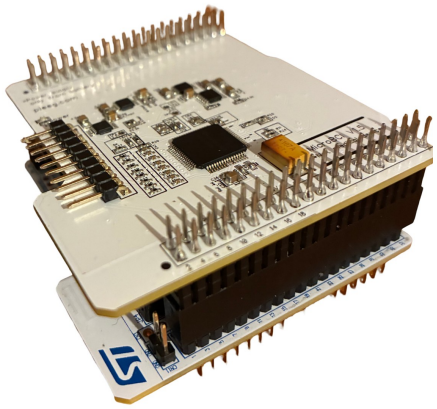
PRODUCT DATASHEET · SKU PIA24024P

8-Channel EEG Shield for STM32 NUCLEO-WB55

A low-cost shield for the STMicroelectronics NUCLEO-WB55 board. A Texas Instruments ADS1299 reads 8 channels of EEG, EMG, ECG or EOG at 24 bits, and the STM32WB55 on the Nucleo streams them over Bluetooth LE to a computer, an Android phone or the browser. Built for embedded and STM32 developers.



8	24-bit	250 SPS	BLE 5	64.5 × 70
Channels + REF + BIAS	TI ADS1299 front end	Default, up to 16 kSPS	STM32WB55 on the Nucleo	mm shield

01 OVERVIEW

Key features

- **Plugs onto the NUCLEO-WB55**: two 2 × 19 sockets on the ST morpho headers — no wiring.
- **Research-grade front end**: TI ADS1299, 8 simultaneous 24-bit channels, programmable gain 1–24.
- **Wireless**: Bluetooth LE with the same protocol as IronBCI — no pairing needed.
- **Open firmware**: STM32CubeIDE project and ready hex; flash over the Nucleo's built-in ST-LINK.
- **Many clients**: Python SDK, Android app, PiEEG Cloud in the browser and PiEEG Server.
- **Open data**: alpha, EMG and EOG recordings with plotting scripts.

Typical applications

- STM32 and embedded-systems courses
- BLE biosignal prototypes
- Alpha, blink and chewing experiments
- EMG and ECG projects
- Firmware development on real biosignals
- Low-cost BCI demonstrators

02 TECHNICAL SPECIFICATIONS

ANALOG ACQUISITION	
Channels	8 biosignal inputs
Extra inputs	Reference (SRB1, shared by all channels) and bias drive
Analog front end	Texas Instruments ADS1299
Resolution	24-bit delta-sigma, all channels sampled at the same moment
Sample rate	250 SPS with the supplied firmware (ADS1299 supports 250 SPS to 16 kSPS)
Gain	ADS1299 supports 1, 2, 4, 6, 8, 12, 24
Reference voltage	4.5 V internal; 1 LSB ≈ 0.268 µV at gain 1
Signals	EEG, EMG, ECG, EOG
HOST & WIRELESS	
Host	STMicroelectronics NUCLEO-WB55 (STM32WB55RG: Cortex-M4 + Cortex-M0+ radio core) — not included
Connectors	2 × 2 × 19 sockets onto ST morpho headers CN7 and CN10

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Bus	SPI1 (SCK PA1, MISO PA6, MOSI PA7), CS PA5, DRDY PA9 · mode 1
Radio	Bluetooth LE 5, advertised name EAREEG, same protocol as IronBCI
Data	24 bytes per sample (8 × 3-byte channels), 250 samples per second
Firmware	MicroBCI hex + STM32WB BLE stack, flashed with STM32CubeProgrammer
Indicators	LED D4 power · LED D3 status
ELECTRODES	
Electrode type	Dry or wet Ag/AgCl
Connector	1 × 10 pin header, 2.54 mm: E1–E8, Biasout, Ref
POWER & SAFETY	
Supply	5 V via the Nucleo's Micro-USB, from a small power bank: ≤ 1 A, ≤ 1,000 mAh
Isolation	None. Battery operation is required
Classification	Research and education tool. Not a medical device
MECHANICAL	
PCB	64.5 × 70.1 mm, 2-layer FR-4, contoured to the NUCLEO-WB55, white mask
Mounting	Held by the morpho sockets; no mounting holes
Shipping weight	100 g (packed)

03 BLOCK DIAGRAM

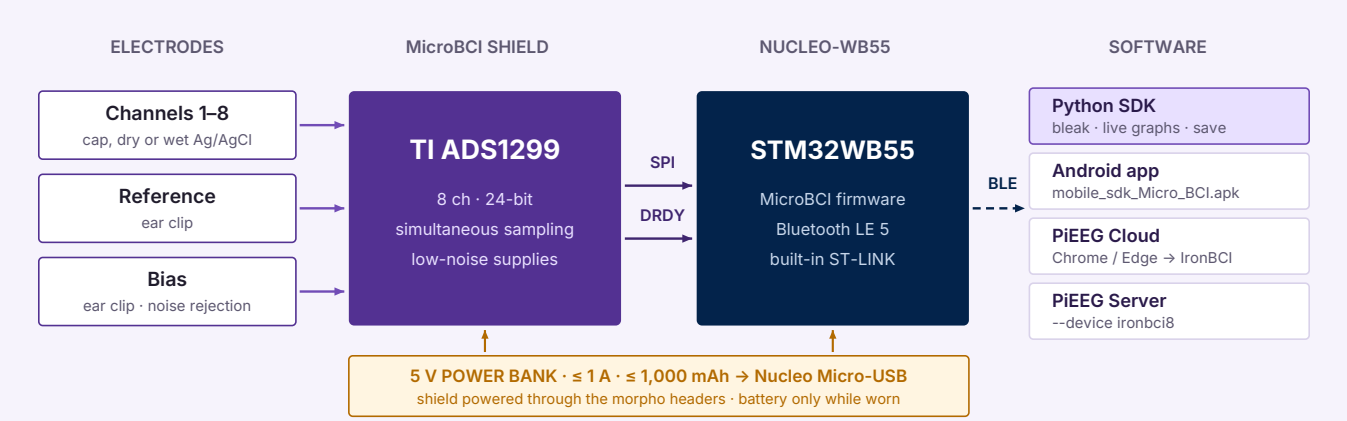


Figure 1 — Signal path. The ADS1299 digitises all channels together; the STM32WB55 reads each sample over SPI when DRDY goes low and sends it over Bluetooth LE.

04 MEASURABLE SIGNALS

Signal	What it shows	Typical first test
EEG	Brain rhythms	Close the eyes and watch alpha (8–12 Hz) rise
EMG	Muscle activity	Chew or clench the jaw
EOG	Eye movement and blinks	Blink with electrodes on the forehead
ECG	Heart activity	Electrodes on the chest or wrists

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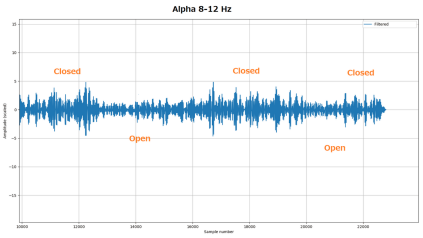


Figure 2 — Alpha test, 8–12 Hz, eyes closed / open.

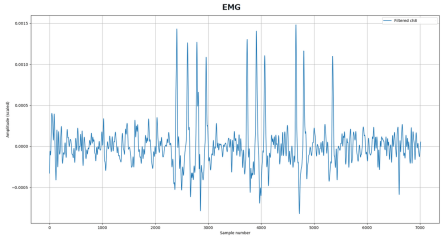


Figure 3 — Chewing (EMG) 4-3-2-1 times.

05 PIEEG 8-CHANNEL FAMILY

	MicroBCI	IronBCI-8	ardEEG	PiEEG-8
Host	**NUCLEO-WB55**	None (built in)	Arduino UNO R4 WiFi	Raspberry Pi
Link	**Bluetooth LE**	Bluetooth LE	Wi-Fi	On the Pi
Firmware	**Flash once**	Pre-installed	Arduino sketch	None
List price	**USD 325**	USD 420	USD 350	USD 350

06 ORDERING INFORMATION

Product	MicroBCI — 8-channel EEG shield for STM32 NUCLEO-WB55
SKU	PIA24024P
Price	USD 325 (list)
Purchase	Elecrow (official manufacturing partner) — elecrow.com/microbci-eeg-with-stm32.html · pieeg.com/microbci

Included	
1 × MicroBCI shield	ADS1299, morpho sockets, electrode header

Not included — required	
NUCLEO-WB55	STM32WB55RG board
Micro-USB data cable	Flashing, then power
5 V power bank	≤ 1 A, ≤ 1,000 mAh
Electrodes and cap	8 Ag/AgCl + 2 ear clips
Receiver	PC, Android or Chrome / Edge

WARNING Battery only.

MicroBCI has no isolation. Flash the Nucleo with no electrodes on anyone; while recording, it runs from a small power bank only — never a computer's USB port or a wall charger. MicroBCI is not a medical device.

07 RELATED DOCUMENTS

MB-MD-002 Mechanical & Dimensional Drawing	**MB-PCB-003** PCB & Manufacturing Specification	**MB-PIN-004** Connector Pinout & Electrode Guide
MB-QSG-005 Quick Start & Safety Guide	**MB-SW-006** Firmware, Bluetooth & Software Guide	

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